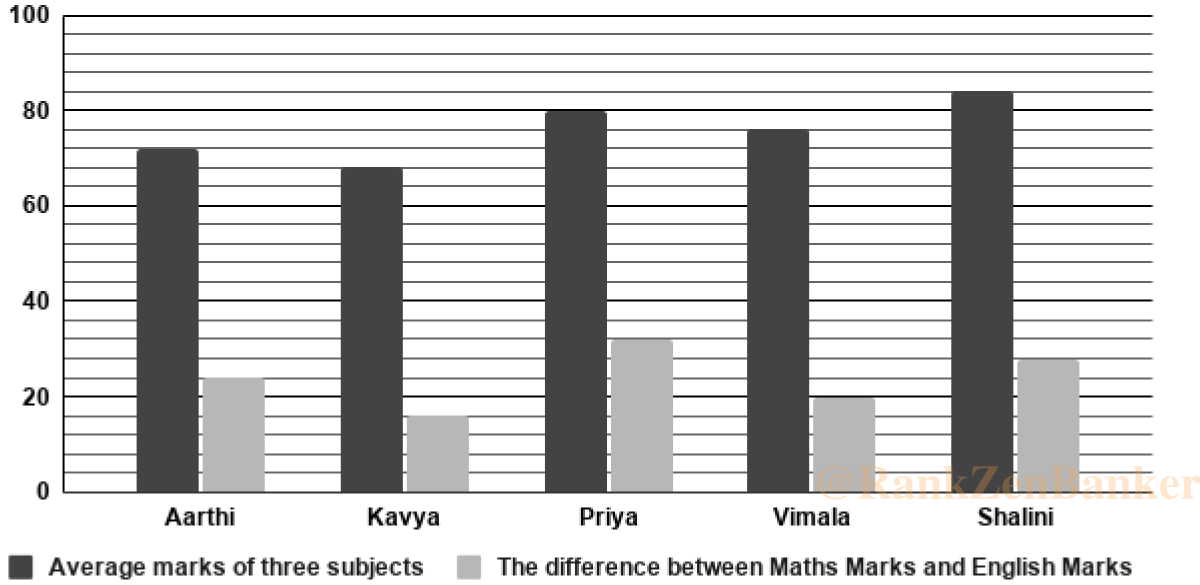
1. Questions

Study the following information carefully and answer the following questions.

The bar graph below shows the average marks of three subjects obtained by five students, namely Aarthi, Kavya, Priya, Vimala, and Shalini, as well as the difference between their Maths and English marks in an examination. Each student appeared for examination in three subjects: Maths, English, and Science.

<u>Note</u>:

For every student, Maths is the highest-scoring subject, and English is the lowest-scoring subject and the marks obtained in Science are equal to the average mark of three subjects shown in the graph.



Find the difference between the average marks scored in Maths by Priya and Kavya together and the average marks scored in Science by Vimala and Shalini together.

a. 8

b. 5

c. 6

d. 9

e. 7

2. Questions

If Aarthi's marks in Maths, English, and Science increased by 25%, 20%, and 50%, respectively, in the next exam, what will be her new total marks in the next exam?

a. 280

b. 285

c. 294

d. 276

e. 290

3. Questions

Find the ratio of the difference between the highest and lowest marks scored in Maths among all five students to the difference between the highest and lowest marks scored in English among all five students.

a. 13 : 6

b. 9 : 4

c. 12 : 5

d. 7 : 3

e. 11 : 5

4. Questions

If the maximum marks are 150, what percentage of the total maximum marks in Maths did Shalini score?

a. 61.25%

b. 42.5%

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c. 72%

d. 65.33%

e. 68%

5. Questions

The highest marks scored in Science among all five students is what percentage of the total marks scored in English by Kavya?

a. 140%

b. 180%

c. 40%

d. 70%

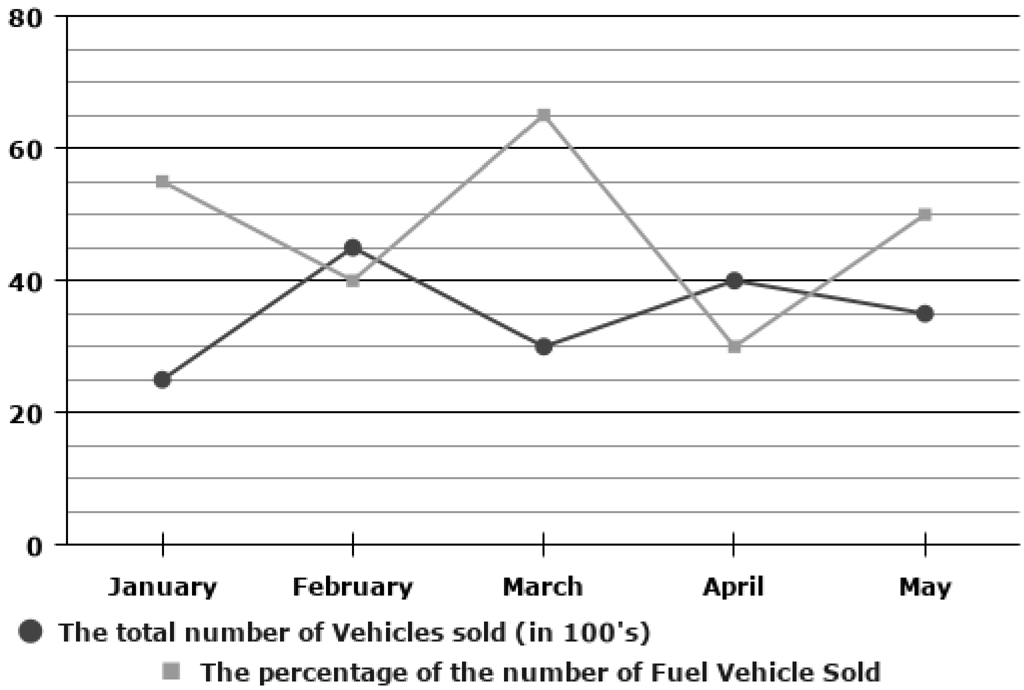
e. 120%

6. Questions

Study the following information carefully and answer the following questions.

A vehicle manufacturing company has sold two different types of Vehicles as Fuel Vehicles and Electric Vehicles. The line graph shows the total number of Vehicles sold and the percentage of the number of Fuel

Vehicles sold out of the total Vehicles sold during the months of January, February, March, April, and May of a year.



The total number of Electric Vehicles sold in January and March together is approximately what percentage of the total number of Electric Vehicles sold in April and May together?

a. 48%

b. 58%

c. 68%

d. 70%

e. 28%

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7. Questions

If 40% of the Electric Vehicles sold in January and 60% of the Electric Vehicles sold in April were two-wheelers, find the total number of Electric Vehicles sold in those two months that were not two-wheelers.

a. 1415

b. 1325

c. 1795

d. 1375

e. 1320

8. Questions

Find the absolute difference between the total number of Fuel Vehicles sold in February and April

together and the total number of Electric Vehicles sold in January and March together.

a. 345

b. 225

c. 925

d. 915

e. 825

9. Questions

If the selling price of a Fuel Vehicle is Rs. 1.5 lakh, and the selling price of an Electric Vehicle is Rs. 1.2 lakh, find the ratio of the total revenue generated from Fuel Vehicles in May to the total revenue generated from Electric Vehicles in April.

a. 27 : 20

b. 25 : 32

c. 21 : 19

d. 25 : 27

e. 19 : 34

10. Questions

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In June, the total vehicles sold is 20% more than that of May. If the ratio of the number of Electric Vehicles sold to Fuel Vehicles sold in June was 7 : 8, find the number of Fuel Vehicles sold in June.

a. 2220

b. 2215

c. 2200

d. 2250

e. 2240

11. Questions

A jar contains a mixture of milk and water in the ratio 4 : 1. If 6 liters of water are added to the jar, the ratio of milk to water becomes 2 : 1. Find the initial quantity of milk in the jar.

a. 9 liters

b. 12 liters

c. 24 liters

d. 6 liters

e. 18 liters

12. Questions

A and B invest in a business in the ratio 4 : 5, respectively. After 6 months, A withdraws half of his capital investment, while B withdraws 20% of his capital investment. Find the ratio of their profit shares at the end of one year.

a. 1 : 3

b. 5 : 3

c. 3 : 7

d. 2 : 3

e. 3 : 4

13. Questions

A sells a watch to B at a profit of 20%, and B sells it to C at a loss of 10%. If C pays Rs. 2,160 for the watch, how much did A originally pay for it?

a. Rs. 2500

b. Rs. 2200

c. Rs. 2450

d. Rs. 2000

e. Rs. 2520

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14. Questions

The length and width of a solid cuboid box are 20 cm and 16 cm, respectively. If its total surface area is 1216 cm^2, find the height of the box.

a. 3 cm

b. 6 cm

c. 5 cm

d. 8 cm

e. 10 cm

15. Questions

A boat covers a distance of 80 km downstream in 2 hours and covers the same distance upstream in 4 hours. Find the time taken by the boat to travel a distance of 45 km in still water.

a. 1.5 hours

b. 1 hour

c. 2.5 hours

d. 2 hours

e. 3 hours

16. Questions

A sum of money invested at simple interest amounts to Rs. 6,400 in 2 years and Rs. 7,600 in 4 years at the same rate of interest. Find the principal sum invested.

a. Rs. 4500

b. Rs. 5000

c. Rs. 5200

d. Rs. 5700

e. Rs. 4000

17. Questions

The average weight of 24 students in a class is 50 kg. If the weight of the teacher is included, the average weight increases by 2 kg. Find the weight of the teacher.

a. 120 kg

b. 100 kg

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c. 96 kg

d. 102 kg

e. 108 kg

18. Questions

A and B can complete a project in 12 days and 18 days, respectively. They start working together, but A leaves the work after 3 days. In how many days will B alone finish the remaining work?

a. 10.5 days

b. 7 days

c. 12 days

d. 11 days

e. 7.5 days

19. Questions

A mother is 20 years older than her daughter. After 5 years, the mother's age will be exactly three times the age of her daughter at that time. Find the present age of the daughter.

a. 1 year

b. 12 years

c. 10 years

d. 8 years

e. 5 years

20. Questions

The present population of a town is 5,000. If it increases at the rate of 12% per annum, what will be its population after 2 years?

a. 6120

b. 6202

c. 6252

d. 6272

e. 6442

21. Questions

What value should come in the place of (?) in the following questions?

35% of 140 + 65% of 180 - √576 = ?

a. 136

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b. 132

c. 142

d. 150

e. 128

22. Questions

(5/9 of 504) + (3/7 of 343) - $(11)^2$ = ?

a. 310

b. 306

c. 315

d. 284

e. 298

23. Questions

(4.5 * 16) + (7.5 * 24) - 360 / 15 = ?

a. 216

b. 228

c. 234

d. 220

e. 240

24. Questions

$\sqrt[3]{4096}$ + 45% of 280 - $(4)^3$ = ?

a. 76

b. 82

c. 78

d. 80

e. 74

25. Questions

(23 * 17) + (19 * 13) - √1024 = ?

a. 606

b. 598

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c. 612

d. 586

e. 600

26. Questions

Find the approximate value that should come in place of the question mark (?) in the following question.

39.98% of 449.92 + 59.91% of 650.18 - $(13.98)^2$ = ?

a. 374

b. 390

c. 365

d. 352

e. 384

27. Questions

√(900.05) * 11.97 / 3.98 + $(14.92)^2$ = ?

a. 305
b. 290
c. 315
d. 325
e. 280

28. Questions

$(12.04)^3 - (8.95)^3 - (5.98)^3 = ?$

a. 783
b. 794
c. 772
d. 801
e. 765

29. Questions

$\sqrt[3]{(2197.03)} * 3.96 + (11.94)^2 / 7.98 = ?$

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a. 48
b. 84
c. 76
d. 60
e. 70

30. Questions

$(3.02)^{? + 2} * (8.97)^4 / (26.98)^2 = (242.95)^{1.03}$

a. 3
b. 2
c. 1
d. 4
e. 0

31. Questions

What value should come in the place of (?) in the following number series?

150, 141, 125, 100, 64, ?

a. 17
b. 15
c. 33
d. 49
e. 36

32. Questions

2, 4, 12, 48, 240, ?

a. 1200
b. 1440
c. 1080
d. 1320
e. 1600

33. Questions

105, 102, 96, 84, 60, ?

a. 12
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b. 18
c. 14
d. 21
e. 11

34. Questions

36, 72, 18, 36, 9, ?

a. 2.5
b. 4.5
c. 24
d. 12
e. 18

35. Questions

11, 12, 26, 81, 328, ?

a. 1610

b. 1635

c. 1655

d. 1645

e. 1640

36. Questions

Find out the wrong number in the following number series.

14, 30, 62, 126, 250, 510

a. 510

b. 126

c. 62

d. 250

e. 30

37. Questions

250, 241, 225, 200, 165, 115

a. 250

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b. 165

c. 200

d. 115

e. 241

38. Questions

8, 24, 20, 60, 56, 158, 164

a. 158

b. 24

c. 56

d. 60

e. 164

39. Questions

20, 22, 25, 30, 37, 48, 60

a. 20

b. 25

c. 60

d. 48

e. 37

40. Questions

5, 9, 17, 26, 53, 68, 133

a. 9

b. 17

c. 53

d. 68

e. 26

41. Questions

In the following questions, two equations numbered I and II are given. Solve both equations and choose the correct answer:

i). $x^2 - 4x - 45 = 0$

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ii). $y^2 - 13y + 40 = 0$

a. $x > y$

b. $x \geq y$

c. $x = y$ or the relationship can't be determined

d. $x < y$

e. $x \leq y$

42. Questions

i). $x^2 - 5x - 36 = 0$

ii). $y^2 - 23y + 132 = 0$

a. $x \leq y$

b. $x < y$

c. $x \geq y$

d. $x = y$, or the relationship can't be determined

e. $x > y$

43. Questions

i). $x^2 - 17x + 72 = 0$

ii). $y^2 - 11y + 24 = 0$

a. $x \geq y$

b. x = y, or the relationship can't be determined

c. $x \leq y$

d. $x > y$

e. $x < y$

44. Questions

i). $x^2 + 5x - 36 = 0$

ii). $y^2 + 19y + 84 = 0$

a. $x \geq y$

b. x = y, or the relationship can't be determined

c. $x \leq y$

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d. $x > y$

e. $x < y$

45. Questions

i). $x^2 - 21x + 108 = 0$

ii). $y^2 - 25y + 156 = 0$

a. $x > y$

b. x = y, or the relationship can't be determined

c. $x < y$

d. $x \geq y$

e. $x \leq y$

Explanations:

1. Questions

Total marks = Average mark of three subjects * 3

Science marks = Average mark of three subjects

Aarthi:

Total marks of Aarthi = 72 * 3 = 216

Science marks of Aarthi = 72

The sum of Maths and English marks of Aarthi = 216 - 72 = 144

Maths marks of Aarthi = [(144 + 24)/2] = 84

English marks of Aarthi = (144 - 24)/2 = 60

Kavya:

Total marks of Kavya = 68 * 3 = 204

Science marks of Kavya = 68

The sum of Maths and English marks of Kavya = 204 - 68 = 136

Maths marks of Kavya = [(136 + 16)/2] = 76

English marks of Kavya = (136 - 16)/2 = 60

Priya:

Total marks of Priya = 80 * 3 = 240

Science marks of Priya = 80

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The sum of Maths and English marks of Priya = 240 - 80 = 160

Maths marks of Priya = [(160 + 32)/2] = 96

English marks of Priya = (160 - 32)/2 = 64

Vimala:

Total marks of Vimala = 76 * 3 = 228

Science marks of Vimala = 76

The sum of Maths and English marks of Vimala = 228 - 76 = 152

Maths marks of Vimala = [(152 + 20)/2] = 86

English marks of Vimala = (152 - 20)/2 = 66

Shalini:

Total marks of Shalini = 84 * 3 = 252

Science marks of Shalini = 84

The sum of Maths and English marks of Shalini = 252 - 84 = 168

Maths marks of Shalini = [(168 + 28)/2] = 98

English marks of Shalini = (168 - 28)/2 = 70

Student	Maths Marks	English Marks	Science Marks	Total marks
Aarthi	84	60	72	216
Kavya	76	60	68	204
Priya	96	64	80	240
Vimala	86	66	76	228
Shalini	98	70	84	252

Answer: C

As per the given information,

The sum of the Maths marks of Priya and Kavya = 76 + 96 = 172

The average Maths marks of Priya and Kavya = 172/2 = 86

The sum of the Science marks of Vimala and Shalini = 76 + 84 = 160

The average Science marks of Vimala and Shalini = 160/2 = 80

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Required difference = 86 - 80 = 6

Hence, the correct answer is option C.

2. Questions

Total marks = Average mark of three subjects * 3

Science marks = Average mark of three subjects

Aarthi:

Total marks of Aarthi = 72 * 3 = 216

Science marks of Aarthi = 72

The sum of Maths and English marks of Aarthi = 216 - 72 = 144

Maths marks of Aarthi = [(144 + 24)/2] = 84

English marks of Aarthi = (144 - 24)/2 = 60

Kavya:

Total marks of Kavya = 68 * 3 = 204

Science marks of Kavya = 68

The sum of Maths and English marks of Kavya = 204 - 68 = 136

Maths marks of Kavya = [(136 + 16)/2] = 76

English marks of Kavya = (136 - 16)/2 = 60

Priya:

Total marks of Priya = 80 * 3 = 240

Science marks of Priya = 80

The sum of Maths and English marks of Priya = 240 - 80 = 160

Maths marks of Priya = [(160 + 32)/2] = 96

English marks of Priya = (160 - 32)/2 = 64

Vimala:

Total marks of Vimala = 76 * 3 = 228

Science marks of Vimala = 76

The sum of Maths and English marks of Vimala = 228 - 76 = 152

Maths marks of Vimala = [(152 + 20)/2] = 86

English marks of Vimala = (152 - 20)/2 = 66

Shalini:

Total marks of Shalini = 84 * 3 = 252

Science marks of Shalini = 84

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The sum of Maths and English marks of Shalini = 252 - 84 = 168

Maths marks of Shalini = [(168 + 28)/2] = 98

English marks of Shalini = (168 - 28)/2 = 70

Student	Maths Marks	English Marks	Science Marks	Total marks
Aarthi	84	60	72	216
Kavya	76	60	68	204
Priya	96	64	80	240
Vimala	86	66	76	228
Shalini	98	70	84	252

Answer: B

As per the given information,

New Maths marks of Aarthi = 84 + [(25/100) * 84] = 84 + 21 = 105

New English marks of Aarthi = 60 + [(20/100) * 60] = 60 + 12 = 72

New Science marks of Aarthi = 72 + [(50/100) * 72] = 72 + 36 = 108

Aarthi's new total marks = 105 + 72 + 108 = 285

Hence, the correct answer is option B.

3. Questions

Total marks = Average mark of three subjects * 3

Science marks = Average mark of three subjects

Aarthi:

Total marks of Aarthi = 72 * 3 = 216

Science marks of Aarthi = 72

The sum of Maths and English marks of Aarthi = 216 - 72 = 144

Maths marks of Aarthi = [(144 + 24)/2] = 84

English marks of Aarthi = (144 - 24)/2 = 60

Kavya:

Total marks of Kavya = 68 * 3 = 204

Science marks of Kavya = 68

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The sum of Maths and English marks of Kavya = 204 - 68 = 136

Maths marks of Kavya = [(136 + 16)/2] = 76

English marks of Kavya = (136 - 16)/2 = 60

Priya:

Total marks of Priya = 80 * 3 = 240

Science marks of Priya = 80

The sum of Maths and English marks of Priya = 240 - 80 = 160

Maths marks of Priya = [(160 + 32)/2] = 96

English marks of Priya = (160 - 32)/2 = 64

Vimala:

Total marks of Vimala = 76 * 3 = 228

Science marks of Vimala = 76

The sum of Maths and English marks of Vimala = 228 - 76 = 152

Maths marks of Vimala = [(152 + 20)/2] = 86

English marks of Vimala = (152 - 20)/2 = 66

Shalini:

Total marks of Shalini = 84 * 3 = 252

Science marks of Shalini = 84

The sum of Maths and English marks of Shalini = 252 - 84 = 168

Maths marks of Shalini = [(168 + 28)/2] = 98

English marks of Shalini = (168 - 28)/2 = 70

Student	Maths Marks	English Marks	Science Marks	Total marks
Aarthi	84	60	72	216
Kavya	76	60	68	204
Priya	96	64	80	240
Vimala	86	66	76	228
Shalini	98	70	84	252

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Answer: E

As per the given information,

The highest marks in Maths = 98

The lowest marks in Maths = 76

The difference between the highest and lowest marks scored in Maths = 98 - 76 = 22

The highest marks in English = 70

The lowest marks in English = 60

The difference between the highest and lowest marks scored in English = 70 - 60 = 10

Required ratio = 22 : 10 = 11 : 5

Hence, the correct answer is option E.

4. Questions

Total marks = Average mark of three subjects * 3

Science marks = Average mark of three subjects

Aarthi:

Total marks of Aarthi = 72 * 3 = 216

Science marks of Aarthi = 72

The sum of Maths and English marks of Aarthi = 216 - 72 = 144

Maths marks of Aarthi = [(144 + 24)/2] = 84

English marks of Aarthi = (144 - 24)/2 = 60

Kavya:

Total marks of Kavya = 68 * 3 = 204

Science marks of Kavya = 68

The sum of Maths and English marks of Kavya = 204 - 68 = 136

Maths marks of Kavya = [(136 + 16)/2] = 76

English marks of Kavya = (136 - 16)/2 = 60

Priya:

Total marks of Priya = 80 * 3 = 240

Science marks of Priya = 80

The sum of Maths and English marks of Priya = 240 - 80 = 160

Maths marks of Priya = [(160 + 32)/2] = 96

English marks of Priya = (160 - 32)/2 = 64

Vimala:

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Total marks of Vimala = 76 * 3 = 228

Science marks of Vimala = 76

The sum of Maths and English marks of Vimala = 228 - 76 = 152

Maths marks of Vimala = [(152 + 20)/2] = 86

English marks of Vimala = (152 - 20)/2 = 66

Shalini:

Total marks of Shalini = 84 * 3 = 252

Science marks of Shalini = 84

The sum of Maths and English marks of Shalini = 252 - 84 = 168

Maths marks of Shalini = [(168 + 28)/2] = 98

English marks of Shalini = (168 - 28)/2 = 70

Student	Maths Marks	English Marks	Science Marks	Total marks
Aarthi	84	60	72	216
Kavya	76	60	68	204
Priya	96	64	80	240
Vimala	86	66	76	228
Shalini	98	70	84	252

Answer: D

As per the given information,

The maximum marks in Maths = 150

The percentage of Shalini's marks in maths = (98/150) * 100 = 196/3 = 65.33 %

Hence, the correct answer is option D.

5. Questions

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Total marks = Average mark of three subjects * 3

Science marks = Average mark of three subjects

Aarthi:

Total marks of Aarthi = 72 * 3 = 216

Science marks of Aarthi = 72

The sum of Maths and English marks of Aarthi = 216 - 72 = 144

Maths marks of Aarthi = [(144 + 24)/2] = 84

English marks of Aarthi = (144 - 24)/2 = 60

Kavya:

Total marks of Kavya = 68 * 3 = 204

Science marks of Kavya = 68

The sum of Maths and English marks of Kavya = 204 - 68 = 136

Maths marks of Kavya = [(136 + 16)/2] = 76

English marks of Kavya = (136 - 16)/2 = 60

Priya:

Total marks of Priya = 80 * 3 = 240

Science marks of Priya = 80

The sum of Maths and English marks of Priya = 240 - 80 = 160

Maths marks of Priya = [(160 + 32)/2] = 96

English marks of Priya = (160 - 32)/2 = 64

Vimala:

Total marks of Vimala = 76 * 3 = 228

Science marks of Vimala = 76

The sum of Maths and English marks of Vimala = 228 - 76 = 152

Maths marks of Vimala = [(152 + 20)/2] = 86

English marks of Vimala = (152 - 20)/2 = 66

Shalini:

Total marks of Shalini = 84 * 3 = 252

Science marks of Shalini = 84

The sum of Maths and English marks of Shalini = 252 - 84 = 168

Maths marks of Shalini = [(168 + 28)/2] = 98

English marks of Shalini = (168 - 28)/2 = 70

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Student	Maths Marks	English Marks	Science Marks	Total marks
Aarthi	84	60	72	216
Kavya	76	60	68	204
Priya	96	64	80	240
Vimala	86	66	76	228
Shalini	98	70	84	252

Answer: A

As per the given information,

Highest marks in Science = 84

Required percentage = (84/60) * 100 = 140%

Hence, the correct answer is option A.

6. Questions

Total number of vehicles sold in January = 2500

The number of Fuel Vehicles sold in January = [(55/100) * 2500] = 1375

The number of Electric Vehicles sold in January = 2500 - 1375 = 1125

Total number of vehicles sold in February = 4500

The number of Fuel Vehicles sold in February = [(40/100) * 4500] = 1800

The number of Electric Vehicles sold in February = 4500 - 1800 = 2700

Total number of vehicles sold in March = 3000

The number of Fuel Vehicles sold in March = [(65/100) * 3000] = 1950

The number of Electric Vehicles sold in March = 3000 - 1950 = 1050

Total number of vehicles sold in April = 4000

The number of Fuel Vehicles sold in April = [(30/100) * 4000] = 1200

The number of Electric Vehicles sold in April = 4000 - 1200 = 2800

Total number of vehicles sold in May = 3500

The number of Fuel Vehicles sold in May = [(50/100) * 3500] = 1750

The number of Electric Vehicles sold in May = 3500 - 1750 = 1750

Month	Total number of vehicles sold	Number of Fuel vehicles sold	Number of Electric vehicles sold
January	2500	1375	1125
February	4500	1800	2700
March	3000	1950	1050
April	4000	1200	2800
May	3500	1750	1750

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Answer: A

As per the given information,

Total number of Electric Vehicles sold in January and March = 1125 + 1050 = 2175

Total number of Electric Vehicles sold in April and May = 2800 + 1750 = 4550

Required percentage = [(2175/4550) * 100] = 47.80% ≈ 48%

Hence, the correct answer is option A.

7. Questions

Total number of vehicles sold in January = 2500

The number of Fuel Vehicles sold in January = [(55/100) * 2500] = 1375

The number of Electric Vehicles sold in January = 2500 - 1375 = 1125

Total number of vehicles sold in February = 4500

The number of Fuel Vehicles sold in February = [(40/100) * 4500] = 1800

The number of Electric Vehicles sold in February = 4500 - 1800 = 2700

Total number of vehicles sold in March = 3000

The number of Fuel Vehicles sold in March = [(65/100) * 3000] = 1950

The number of Electric Vehicles sold in March = 3000 - 1950 = 1050

Total number of vehicles sold in April = 4000

The number of Fuel Vehicles sold in April = [(30/100) * 4000] = 1200

The number of Electric Vehicles sold in April = 4000 - 1200 = 2800

Total number of vehicles sold in May = 3500

The number of Fuel Vehicles sold in May = [(50/100) * 3500] = 1750

The number of Electric Vehicles sold in May = 3500 - 1750 = 1750

Month	Total number of vehicles sold	Number of Fuel vehicles sold	Number of Electric vehicles sold
January	2500	1375	1125
February	4500	1800	2700
March	3000	1950	1050
April	4000	1200	2800
May	3500	1750	1750

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Answer: C

As per the given information,

The number of Electric Vehicles that were not two-wheelers in January = [(60/100) * 1125] = 675

The number of Electric Vehicles that were not two-wheelers in April = [(40/100) * 2800] = 1120

Required total non-two-wheeler Electric Vehicles = 675 + 1120 = 1795

Hence, the correct answer is option C.

8. Questions

Total number of vehicles sold in January = 2500

The number of Fuel Vehicles sold in January = [(55/100) * 2500] = 1375

The number of Electric Vehicles sold in January = 2500 - 1375 = 1125

Total number of vehicles sold in February = 4500

The number of Fuel Vehicles sold in February = [(40/100) * 4500] = 1800

The number of Electric Vehicles sold in February = 4500 - 1800 = 2700

Total number of vehicles sold in March = 3000

The number of Fuel Vehicles sold in March = [(65/100) * 3000] = 1950

The number of Electric Vehicles sold in March = 3000 - 1950 = 1050

Total number of vehicles sold in April = 4000

The number of Fuel Vehicles sold in April = [(30/100) * 4000] = 1200

The number of Electric Vehicles sold in April = 4000 - 1200 = 2800

Total number of vehicles sold in May = 3500

The number of Fuel Vehicles sold in May = [(50/100) * 3500] = 1750

The number of Electric Vehicles sold in May = 3500 - 1750 = 1750

Month	Total number of vehicles sold	Number of Fuel vehicles sold	Number of Electric vehicles sold
January	2500	1375	1125
February	4500	1800	2700
March	3000	1950	1050
April	4000	1200	2800
May	3500	1750	1750

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Answer: E

As per the given information,

The total number of Fuel Vehicles sold in February and April = 1800 + 1200 = 3000

The total number of Electric Vehicles sold in January and March = 1125 + 1050 = 2175

The required difference = 3000 - 2175 = 825

Hence, the correct answer is option E.

9. Questions

Total number of vehicles sold in January = 2500

The number of Fuel Vehicles sold in January = [(55/100) * 2500] = 1375

The number of Electric Vehicles sold in January = 2500 - 1375 = 1125

Total number of vehicles sold in February = 4500

The number of Fuel Vehicles sold in February = [(40/100) * 4500] = 1800

The number of Electric Vehicles sold in February = 4500 - 1800 = 2700

Total number of vehicles sold in March = 3000

The number of Fuel Vehicles sold in March = [(65/100) * 3000] = 1950

The number of Electric Vehicles sold in March = 3000 - 1950 = 1050

Total number of vehicles sold in April = 4000

The number of Fuel Vehicles sold in April = [(30/100) * 4000] = 1200

The number of Electric Vehicles sold in April = 4000 - 1200 = 2800

Total number of vehicles sold in May = 3500

The number of Fuel Vehicles sold in May = [(50/100) * 3500] = 1750

The number of Electric Vehicles sold in May = 3500 - 1750 = 1750

Month	Total number of vehicles sold	Number of Fuel vehicles sold	Number of Electric vehicles sold
January	2500	1375	1125
February	4500	1800	2700
March	3000	1950	1050
April	4000	1200	2800
May	3500	1750	1750

Answer: B

As per the given information,

The total revenue generated from Fuel Vehicles in May = 1750 * 1.5 = 2625 lakhs

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The total revenue generated from Electric Vehicles in April = 2800 * 1.2 = 3360 lakhs

Required ratio = 2625 : 3360 = 25 : 32

Hence, the correct answer is option B.

10. Questions

Total number of vehicles sold in January = 2500

The number of Fuel Vehicles sold in January = [(55/100) * 2500] = 1375

The number of Electric Vehicles sold in January = 2500 - 1375 = 1125

Total number of vehicles sold in February = 4500

The number of Fuel Vehicles sold in February = [(40/100) * 4500] = 1800

The number of Electric Vehicles sold in February = 4500 - 1800 = 2700

Total number of vehicles sold in March = 3000

The number of Fuel Vehicles sold in March = [(65/100) * 3000] = 1950

The number of Electric Vehicles sold in March = 3000 - 1950 = 1050

Total number of vehicles sold in April = 4000

The number of Fuel Vehicles sold in April = [(30/100) * 4000] = 1200

The number of Electric Vehicles sold in April = 4000 - 1200 = 2800

Total number of vehicles sold in May = 3500

The number of Fuel Vehicles sold in May = [(50/100) * 3500] = 1750

The number of Electric Vehicles sold in May = 3500 - 1750 = 1750

Month	Total number of vehicles sold	Number of Fuel vehicles sold	Number of Electric vehicles sold
January	2500	1375	1125
February	4500	1800	2700
March	3000	1950	1050
April	4000	1200	2800
May	3500	1750	1750

Answer: E

As per the given information,

The total number of overall vehicles sold in June = 3500 + [(20/100) * 3500] = 3500 + 700 = 4200

The number of Fuel Vehicles sold in June = [(4200/15) * 8] = (8 * 280) = 2240

Hence, the correct answer is option E.

11. Questions

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Answer: C

As per the given information,

Let the initial quantities of milk be 4x, and water be 1x.

When 6 liters of water are added, the quantity of milk remains unchanged.

[4x/(x + 6)] = 2/1

4x = 2x + 12

2x = 12

x = 6

The initial quantity of milk in the jar = 4 * 6 = 24 liters

Hence, the correct answer is option C.

12. Questions

Answer: D

As per the given information,

Let the initial investments of A and B be 4x and 5x

A’s investment after 6 months = 4x/2 = 2x

B’s investment after 6 months = [5x - {(20/100) * 5x}] = 5x - x = 4x

Profit ratio of A and B = [(6 * 4x) + (6 * 2x)] : [(6 * 5x) + (6 * 4x)]

= (24x + 12x) : (30x + 24x) = 36x : 54x = 2 : 3

Hence, the correct answer is option D.

13. Questions

Answer: D

As per the given information,

Let the original price paid by A be 100x

The price paid by B = 100x + [20% of 100x] = 120x

The price paid by C = 120x - [10% of 120x] = 120x - 12x = 108x

Given,

108x = 2160

x = 20

The original price paid by A = 100 * 20 = Rs. 2000

Hence, the correct answer is option D.

14. Questions

Answer: D

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As per the given information,

Total surface area of the cuboid = [2 * (lw + wh + hl)] = 1216 cm^2

l = 20, w = 16, h = ?

lw + wh + hl = 1216/2

(20 * 16) + 16h + 20h = 608

320 + 36h = 608

36h = 288

h = 8 cm

The height of the cuboid box = 8 cm

Hence, the correct answer is option D.

15. Questions

Answer: A

As per the given information,

Downstream speed = 80/2 = 40 km/hr

Upstream speed = 80/4 = 20 km/hr

Speed of the boat in still water = [(40 + 20) / 2] = 30 km/hr

Time to cover 45 km in still water = 45/30 = 1.5 hours

Hence, the correct answer is option A.

16. Questions

Answer: C

As per the given information,

Amount in 4 years - Amount in 2 years = Simple interest in 2 years

7600 - 6400 = 1200

Total amount at two years = Principal + Simple interest for 2 years

6400 = Principal + 1200

Principal = 6400 - 1200 = Rs. 5200

Hence, the correct answer is option C.

17. Questions

Answer: B

As per the given information,

Initial total weight of the class = 24 * 50 = 1200 kg

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The new total number of people becomes = 24 + 1 = 25

The new average weight = 50 + 2 = 52 kg

The new total weight of the class = 25 * 52 = 1300 kg

The weight of the teacher = 1300 - 1200 = 100 kg

Hence, the correct answer is option B.

18. Questions

Answer: A

As per the given information,

Work done by A and B in 3 days = [(1/12) + (1/18)] * 3 = [(3 + 2)/36] * 3 = 5/12

Remaining work = 1 - 5/12 = 7/12

B alone finishes the remaining work = [(7/12) * 18] = 21/2 = 10.5 days

Hence, the correct answer is option A.

19. Questions

Answer: E

As per the given information,

After 5 years, the age ratio of a mother and her daughter = 3 : 1

Age difference = 20

Daughter's age after 5 years = (20/2) * 1 = 10 years

The present age of the daughter = 10 - 5 = 5 years

Hence, the correct answer is option E.

20. Questions

Answer: D

As per the given information,

The new population at the end of one year = 5000 + [(12/100) * 5000] = 5000 + 600 = 5600

The new population at the end of second year = 5600 + [(12/100) * 5600] = 5600 + 672 = 6272

Hence, the correct answer is option D.

21. Questions

Answer: C

35% of 140 + 65% of 180 - √576 = ?

[(35/100) * 140] + [(65/100) * 180] - 24 = ?

49 + 117 - 24 = ?

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166 - 24 = ?

142 = ?

Hence, the correct answer is option C.

22. Questions

Answer: B

(5/9 of 504) + (3/7 of 343) - $(11)^2$ =?

5 * 56 + 3 * 49 - 121 = ?

280 + 147 - 121 = ?

306 = ?

Hence, the correct answer is option B.

23. Questions

Answer: B

(4.5 * 16) + (7.5 * 24) - 360 / 15 = ?

72 + 180 - 24 = ?

252 - 24 =?

228 = ?

Hence, the correct answer is option B.

24. Questions

Answer: C

$\sqrt[3]{4096}$ + 45% of 280 - $(4)^3$ = ?

16 + [(45/100) * 280] - 64 = ?

16 + 126 - 64 =?

142 - 64 = ?

78 = ?

Hence, the correct answer is option C.

25. Questions

Answer: A

(23 * 17) + (19 * 13) - √1024 = ?

391 + 247 - 32 = ?

638 - 32 = ?

606 = ?

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Hence, the correct answer is option A.

26. Questions

Answer: A

39.98% of 449.92 + 59.91% of 650.18 - $(13.98)^2$ = ?

40% of 450 + 60% of 650 - $(14)^2$ = ?

180 + 390 - 196 = ?

570 - 196 = ?

374 = ?

Hence, the correct answer is option A.

27. Questions

Answer: C

√(900.05) * 11.97 / 3.98 + $(14.92)^2$ = ?

√(900) * 12 / 4 + $(15)^2$ = ?

(30 * 3) + 225 = ?

90 + 225 = ?

315 = ?

Hence, the correct answer is option C.

28. Questions

Answer: A

$(12.04)^3 - (8.95)^3 - (5.98)^3 = ?$

$(12)^3 - (9)^3 - (6)^3 = ?$

1728 - 729 - 216 = ?

1728 - 945 = ?

783 = ?

Hence, the correct answer is option A.

29. Questions

Answer: E

$\sqrt[3]{(2197.03)} * 3.96 + (11.94)^2 / 7.98 = ?$

$\sqrt[3]{(2197)} * 4 + (12)^2 / 8 = ?$

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13 * 4 + 144/8 = ?

52 + 18 = ?

70 = ?

Hence, the correct answer is option E.

30. Questions

Answer: C

$(3.02)^{?+2} * (8.97)^4 / (26.98)^2 = (242.95)^{1.03}$

$(3)^{?+2} * (9)^4 / (27)^2 = (243)^1$

$(3)^{?+2} * (3^2)^4 / (3^3)^2 = (3)^5$

$(3)^{?+2} * (3^8 / 3^6) = (3)^5$

$3^{(?+2+8-6)} = (3)^5$

$3^{?+4} = 3^5$

? + 4 = 5

? = 1

Hence, the correct answer is option C.

31. Questions

Answer: B

150, 141, 125, 100, 64, ?

$150 - 3^2 = 141$

$141 - 4^2 = 125$

$125 - 5^2 = 100$

$100 - 6^2 = 64$

$64 - 7^2 = \mathbf{15}$

Hence, the correct answer is option B.

32. Questions

Answer: B

2, 4, 12, 48, 240, ?

2 * 2 = 4

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4 * 3 = 12

12 * 4 = 48

48 * 5 = 240

240 * 6 = **1440**

Hence, the correct answer is option B.

33. Questions

Answer: A

105 - 3 = 102

102 - 6 = 96

96 - 12 = 84

84 - 24 = 60

60 - 48 = **12**

Hence, the correct answer is option A.

34. Questions

Answer: E

36, 72, 18, 36, 9, ?

36 * 2 = 72

72 / 4 = 18

18 * 2 = 36

36 / 4 = 9

9 * 2 = **18**

Hence, the correct answer is option E.

35. Questions

Answer: D

11, 12, 26, 81, 328, ?

[(11 * 1) + 1] = 12

[(12 * 2) + 2] = 26

[(26 * 3) + 3] = 81

[(81 * 4) + 4] = 328

[(328 * 5) + 5] = **1645**

Hence, the correct answer is option D.

36. Questions

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Answer: D

14, 30, 62, 126, 250, 510

14 * 2 + 2 = 30

30 * 2 + 2 = 62

62 * 2 + 2 = 126

126 * 2 + 2 = **254**

254 * 2 + 2 = 510

Hence, the correct answer is option D.

37. Questions

Answer: B

250, 241, 225, 200, 165, 115

250 - 9 = 241

241 - 16 = 225

225 - 25 = 200

200 - 36 = **164**

164 - 49 = 115

Hence, the correct answer is option B.

38. Questions

Answer: A

8, 24, 20, 60, 56, 158, 164

8 * 3 = 24

24 - 4 = 20

20 * 3 = 60

60 - 4 = 56

56 * 3 = **168**

168 - 4 = 164

Hence, the correct answer is option A.

39. Questions

Answer: C

20, 22, 25, 30, 37, 48, 60

20 + 2 = 22

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22 + 3 = 25

25 + 5 = 30

30 + 7 = 37

37 + 11 = 48

48 + 13 = **61**

Hence, the correct answer is option C.

40. Questions

Answer: D

5, 9, 17, 26, 53, 68

5 + 4 (2^2) = 9

9 + 8 (2^3) = 17

17 + 9 (3^2) = 26

26 + 27 (3^3) = 53

53 + 16 (4^2) = **69**

$69 + 64\ (4^3) = 133$

Hence, the correct answer is option D.

41. Questions

Answer: C

i). $x^2 - 4x - 45 = 0$

$x^2 - 9x + 5x - 45 = 0$

$x(x - 9) + 5(x - 9) = 0$

$(x + 5)\ (x - 9) = 0$

$x = -5, 9$

ii). $y^2 - 13y + 40 = 0$

$y^2 - 8y - 5y + 40 = 0$

$y(y - 8) - 5(y - 8) = 0$

$(y - 5)\ (y - 8) = 0$

$y = 5, 8$

While comparing every x value with every y value, a relationship cannot be established.

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Hence, the correct answer is option C.

42. Questions

Answer: B

i). $x^2 - 5x - 36 = 0$

$x^2 - 9x + 4x - 36 = 0$

$x(x - 9) + 4(x - 9) = 0$

$(x - 9)\ (x + 4) = 0$

$x = 9, -4$

ii). $y^2 - 23y + 132 = 0$

$y^2 - 11y - 12y + 132 = 0$

$y(y - 11) - 12(y - 11) = 0$

$(y - 11)\ (y - 12) = 0$

$y = 11, 12$

While comparing every x value with every y value, all x values are less than y. So, $x < y$

Hence, the correct answer is option B.

43. Questions

Answer: A

i). $x^2 - 17x + 72 = 0$

$x^2 - 8x - 9x + 72 = 0$

$x(x - 8) - 9(x - 8) = 0$

$(x - 9)(x - 8) = 0$

$x = 9, 8$

ii). $y^2 - 11y + 24 = 0$

$y^2 - 8y - 3y + 24 = 0$

$y(y - 8) - 3(y - 8) = 0$

$(y - 8)(y - 3) = 0$

$y = 8, 3$

While comparing every x value with every y value, $x \geq y$

Hence, the correct answer is option A.

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44. Questions

Answer: B

i). $x^2 + 5x - 36 = 0$

$x^2 + 9x - 4x - 36 = 0$

$x(x + 9) - 4(x + 9) = 0$

$(x + 9)(x - 4) = 0$

$x = -9, 4$

ii). $y^2 + 19y + 84 = 0$

$y^2 + 12y + 7y + 84 = 0$

$y(y + 12) + 7(y + 12) = 0$

$(y + 12)(y + 7) = 0$

$y = -12, -7$

While comparing every x value with every y value, a relationship cannot be established.

Hence, the correct answer is option B.

45. Questions

Answer: E

i). $x^2 - 21x + 108 = 0$

$x^2 - 12x - 9x + 108 = 0$

$x(x - 12) - 9(x - 12) = 0$

$(x - 9)(x - 12) = 0$

$x = 9, 12$

ii). $y^2 - 25y + 156 = 0$

$y^2 - 12y - 13y + 156 = 0$

$y(y - 12) - 13(y - 12) = 0$

$(y - 12)(y - 13) = 0$

$y = 12, 13$

While comparing every x value with every y value, $x \leq y$

Hence, the correct answer is option E.

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